AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 20 · STUDENT LESSON

Repair Bias in Data

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Combine the counts. Inspect who entered them. Repair the process.

Separate respondent arithmetic, weighting assumptions, collection changes and causal explanations.

Keep one habit: Group counts → unequal respondent mix → separate weighting illustration → documented repair and limits.

MAIN LESSON · SECTION 1

Does a bigger sample fix a restricted collection process?

> Combine observed counts correctly and keep the population claim separate.

Your learning goals

I can identify selection and coverage **bias**.

I can compare groups before combining results.

I can explain why more **bias**ed data may not help.

Recall population versus sample

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- What is the difference between a population and a sample?

Your end goal

By the end, I can distinguish a **bias**ed collection **process** from a mere small dataset.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Sampling and Representation. Give an example and a mistake that the idea helps you avoid.

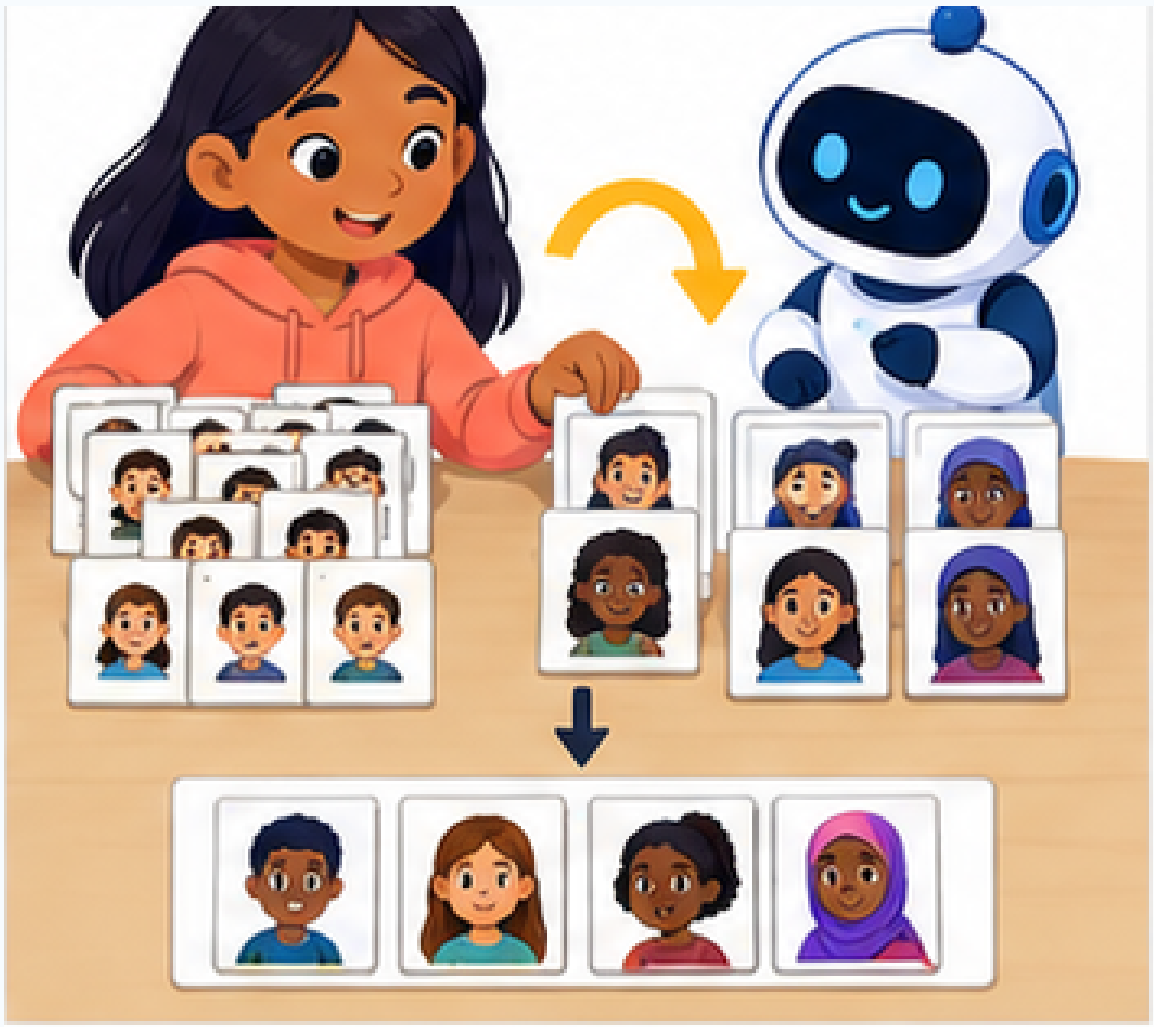
Without opening the old lesson, explain one key idea from Noise and Outliers. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 20: Repair **Bias** in Data

Fairness work changes data, features, tests, or use conditions, then rechecks outcomes.
</details>



THINKER MISSION

Redesign an imbalanced speech dataset and specify **retest**s.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Start with two groups and their observed replies

> The collection process determines who can enter the data.

The complete opening poll counts

A fictional poll has 8 Yes among 10 club respondents and 2 Yes among 10 other respondents. If only club members are surveyed, an 80 percent result misses a group with a different pattern.

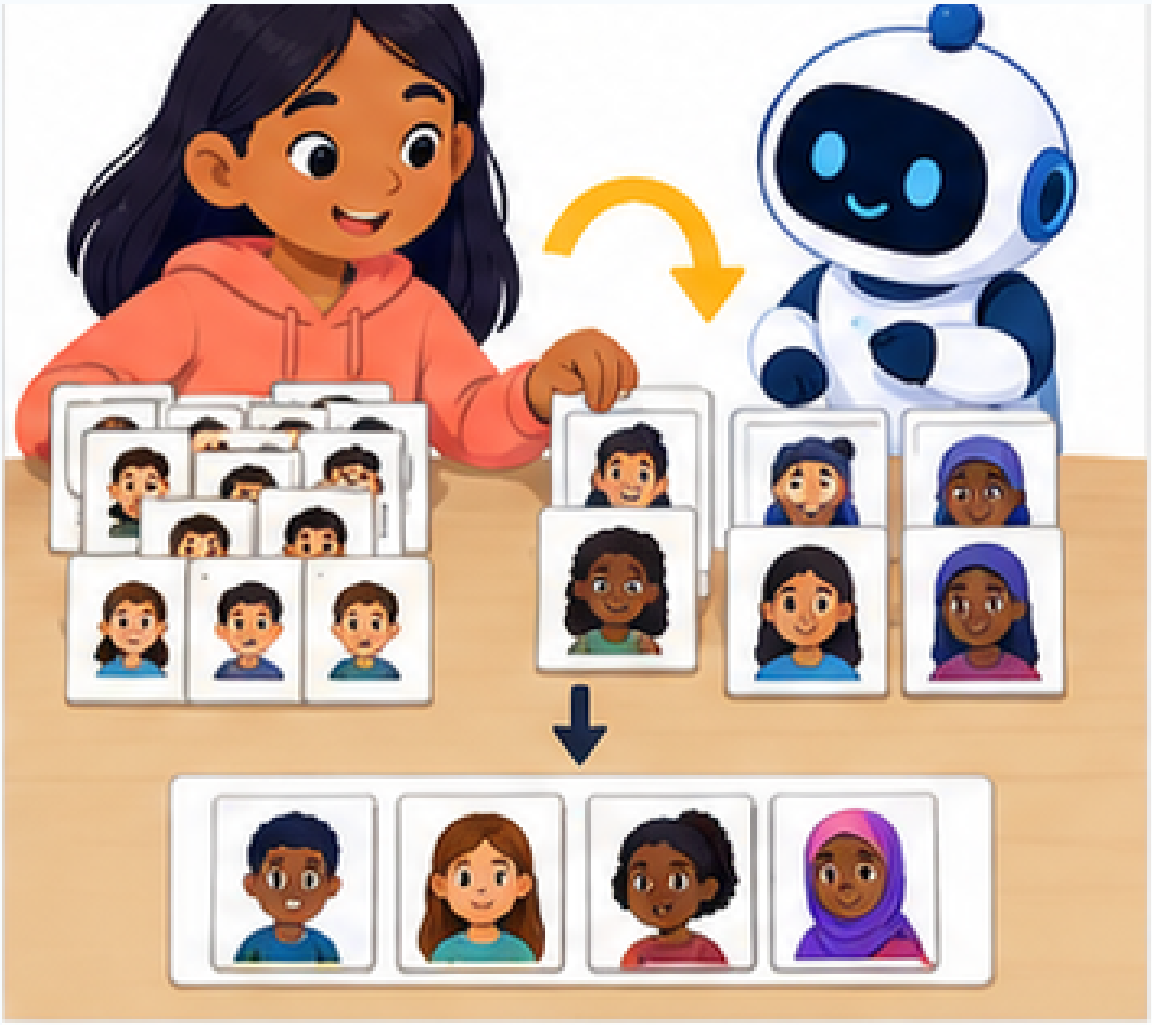
Examine who is included before treating a larger sample as better.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use repair **bias** in data to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 3

Combine numerators and denominators before dividing

> A correct respondent rate is not automatically a population estimate.

Predict and execute the complete count script

```
club_yes, club_n = 8, 10
other_yes, other_n = 2, 10
print(club_yes / club_n)
print(other_yes / other_n)
print((club_yes + other_yes) / (club_n + other_n))
```

Read each group and the combined sample

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Club sample	8 of 10	80 percent
Other sample	2 of 10	20 percent
Combined sample	10 of 20	50 percent

Read each row from left to right and connect the evidence to the stated rule.

The combined observed sample has 50 percent Yes. This is not automatically a population estimate: we also need to know how the groups were selected and how many people each represents.

Do not average percentages without considering their denominators.

Explain why group size matters

The club has 8 Yes out of 10, and the other group has 2 Yes out of 10. Combine the Yes counts to get 10 and combine the response counts to get 20; the combined observed rate is 10 divided by 20, or 50 percent.

When group sizes differ, simply averaging their percentages can give the wrong combined rate. A correct combined calculation still describes the collected sample, not automatically the whole school.

Point to the deciding denominator

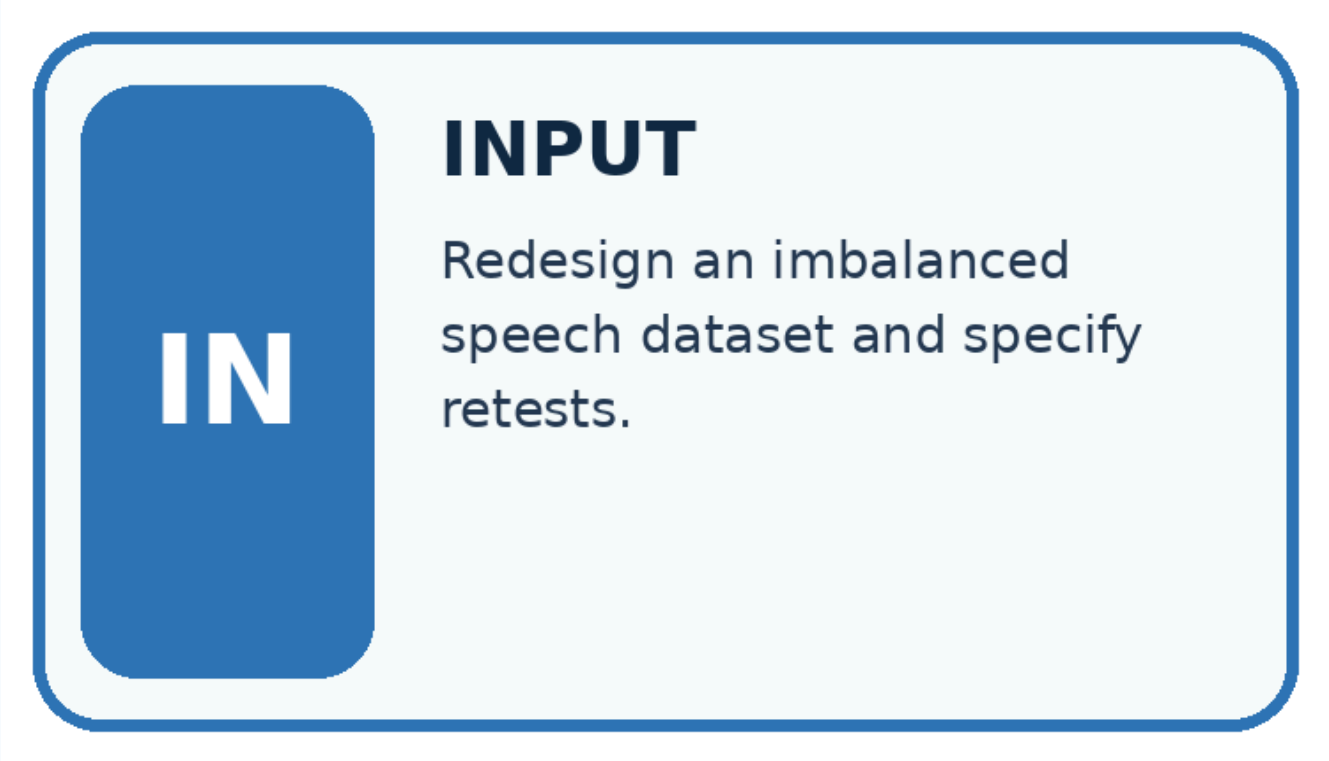
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

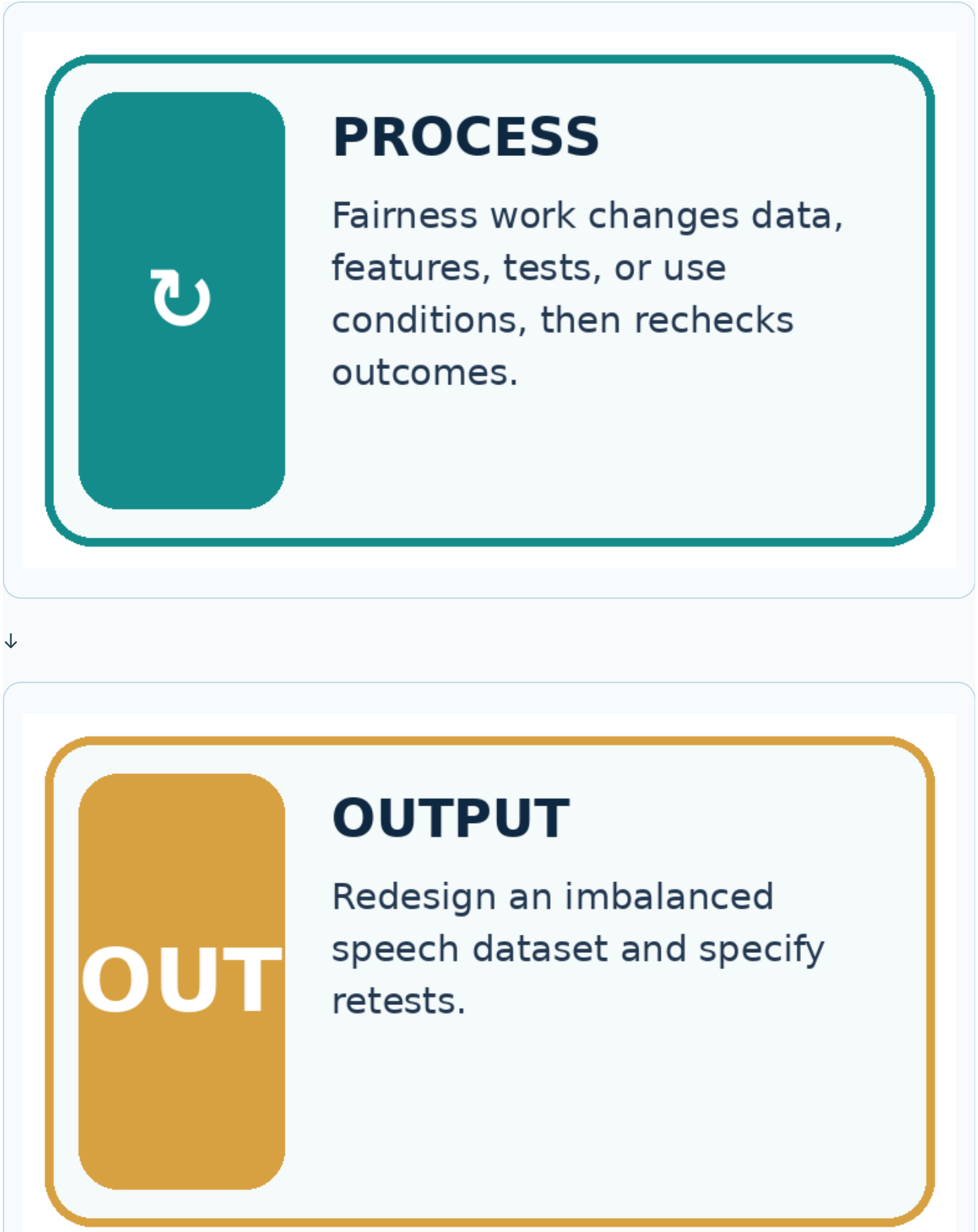
VISUAL **MECHANISM**

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓


PROCESS
Fairness work changes data, features, tests, or use conditions, then rechecks outcomes.
↓
OUT
OUTPUT
Redesign an imbalanced speech dataset and specify retests.

Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

Combine counts, not just percentages

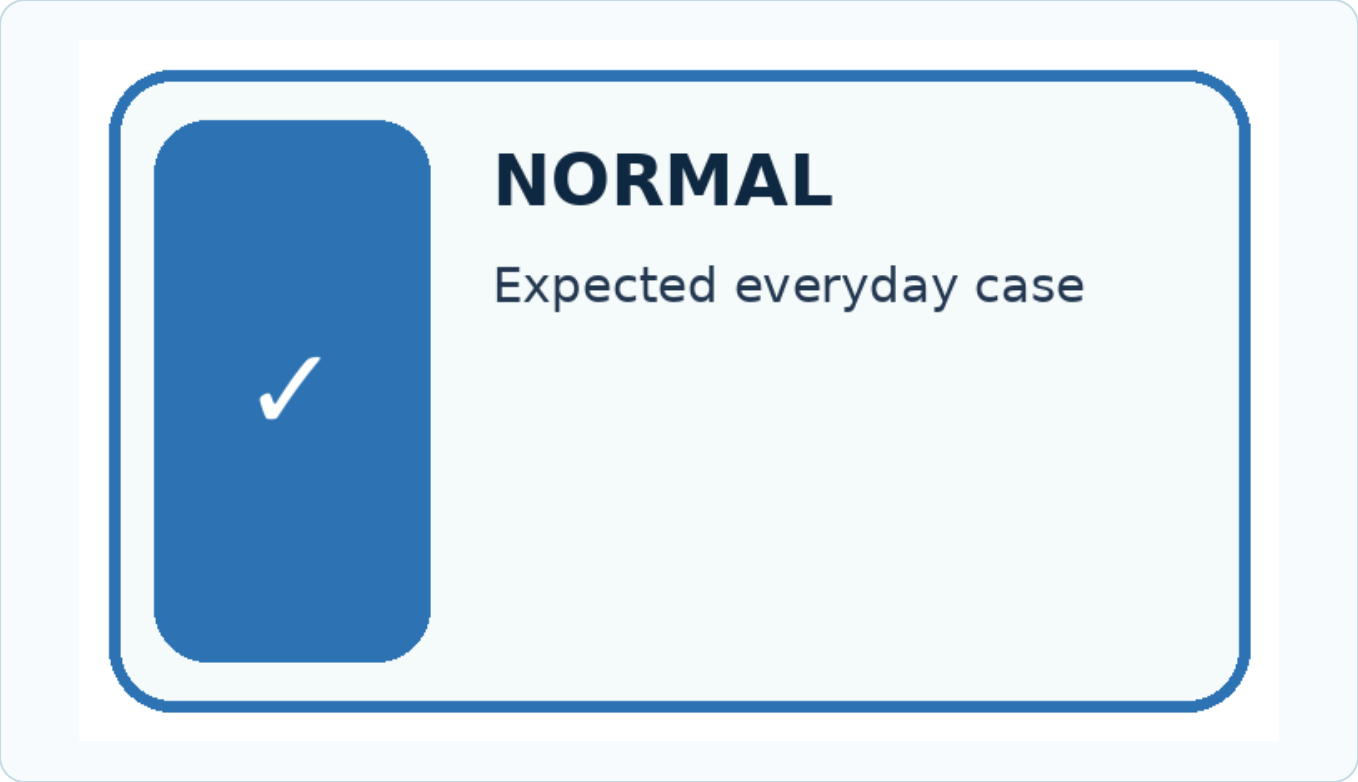
Connect the picture and the explanation

MAIN LESSON · SECTION 4

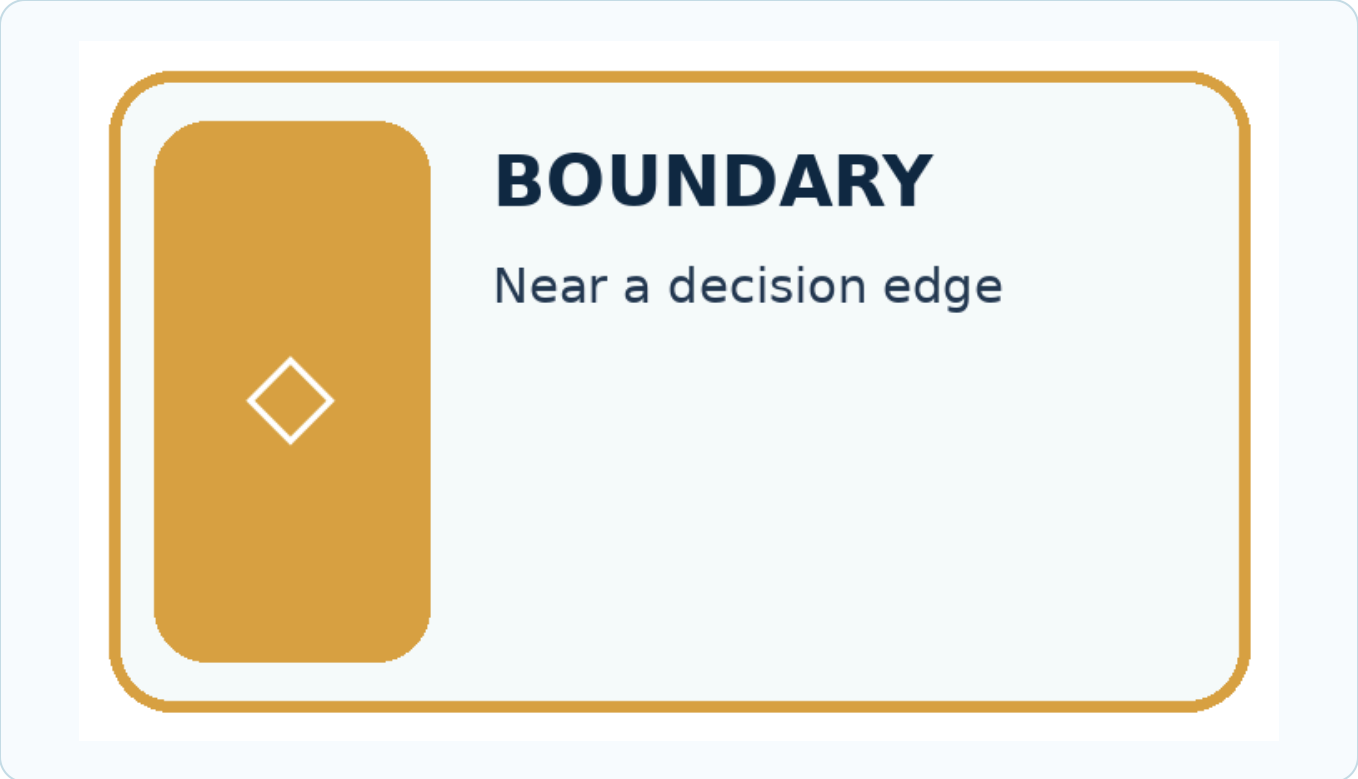
Choose a repair that changes coverage

> Copying rows or adding the same group is not new diverse evidence.

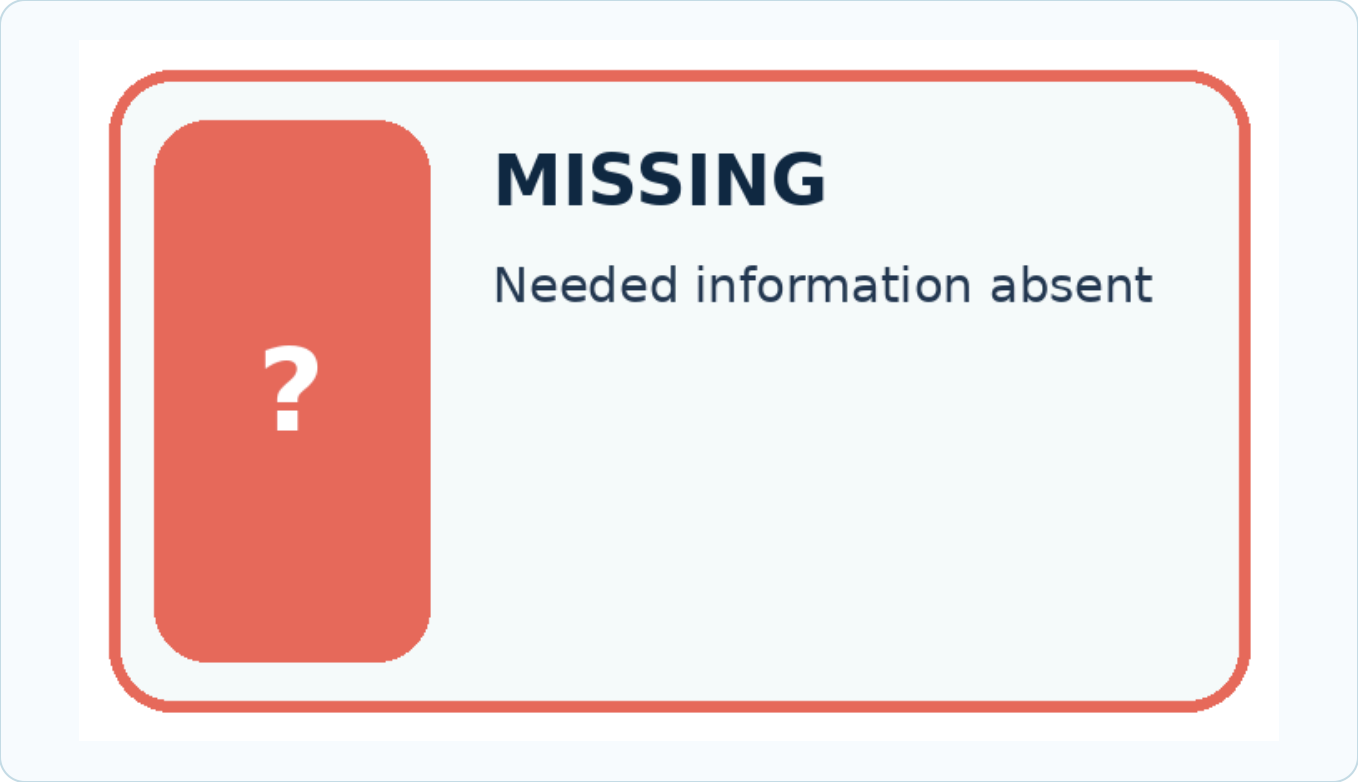
Four pictures with a purpose



Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Predict the supplied changed case

- Change: Recruit more club volunteers only.

▼ Check after predicting

Prediction: Sample size increases without fixing missing representation of other learners.

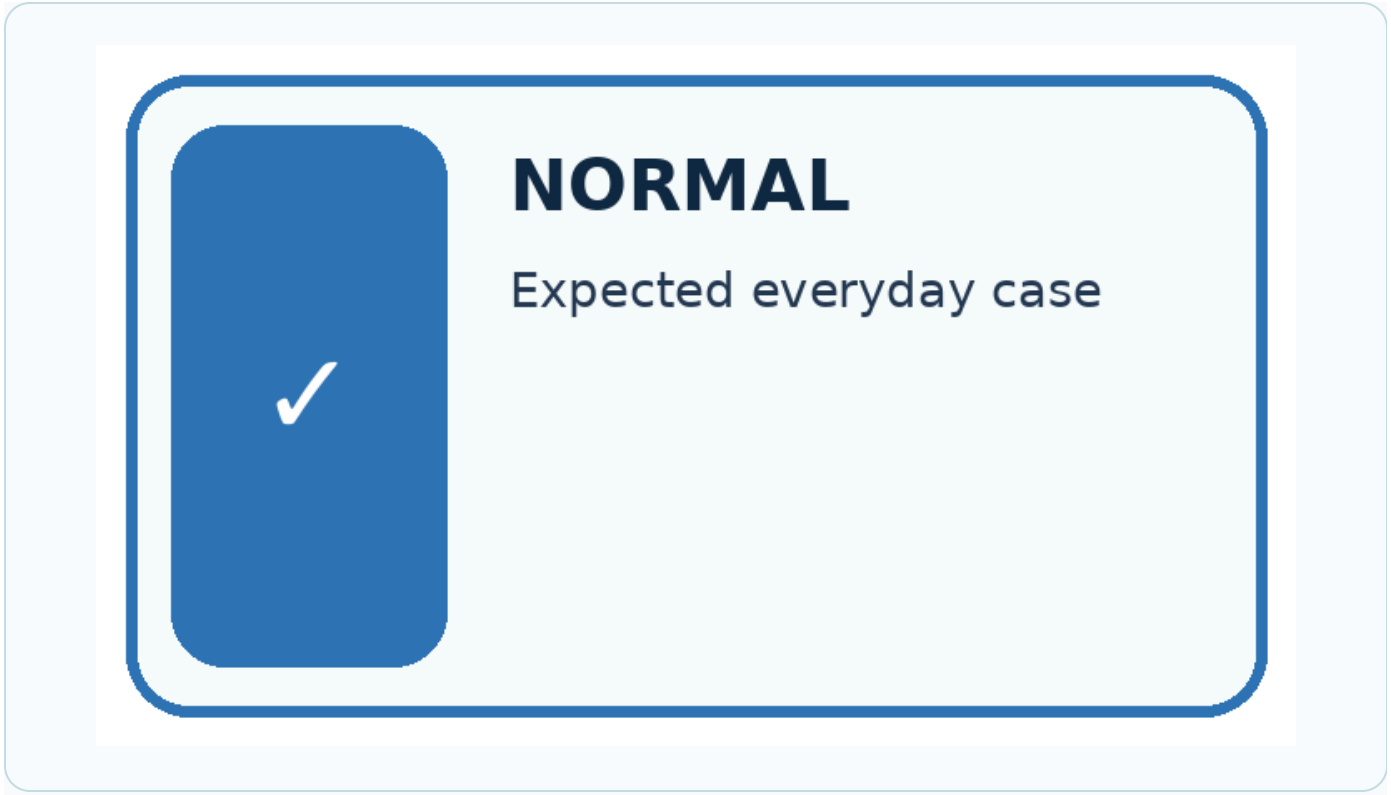
Reason: The added respondents come from the already overrepresented group.

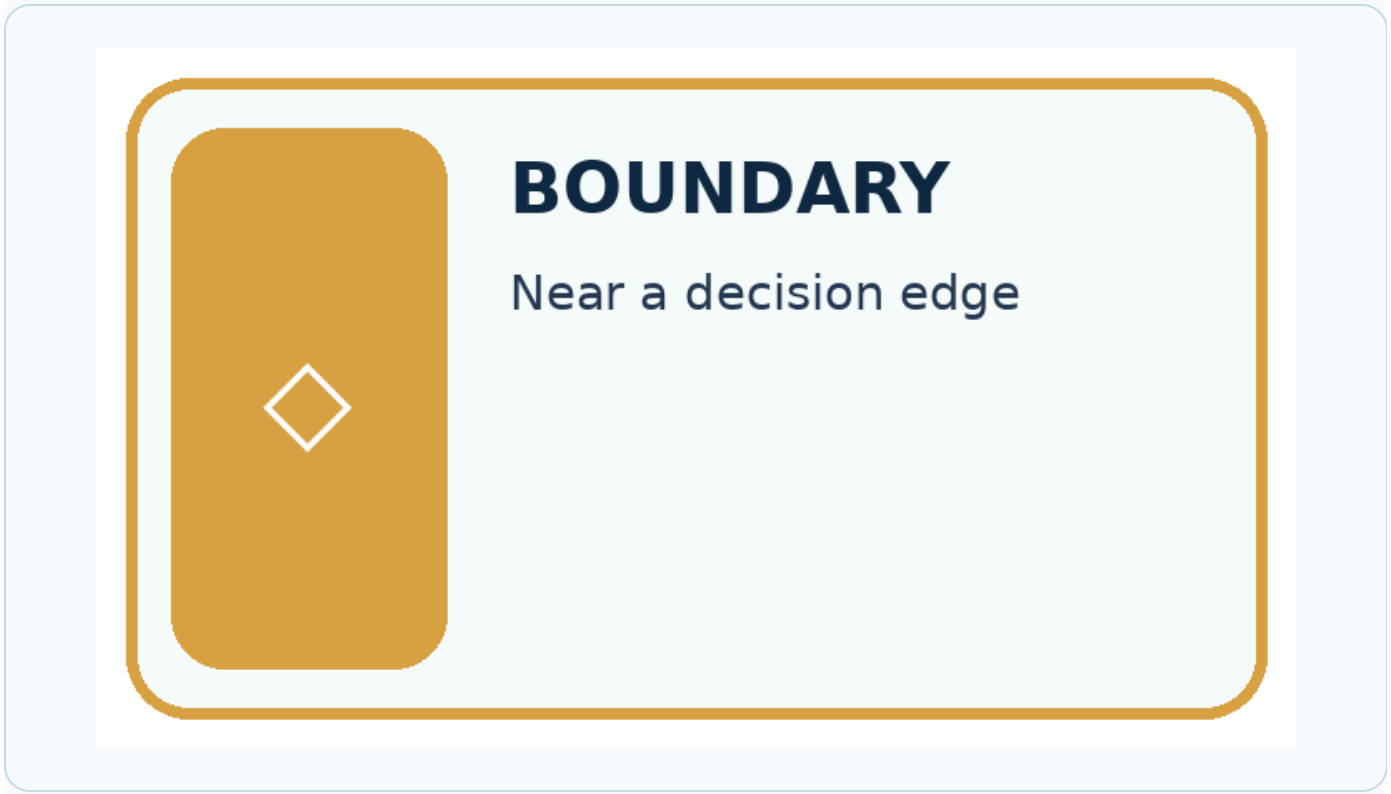
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

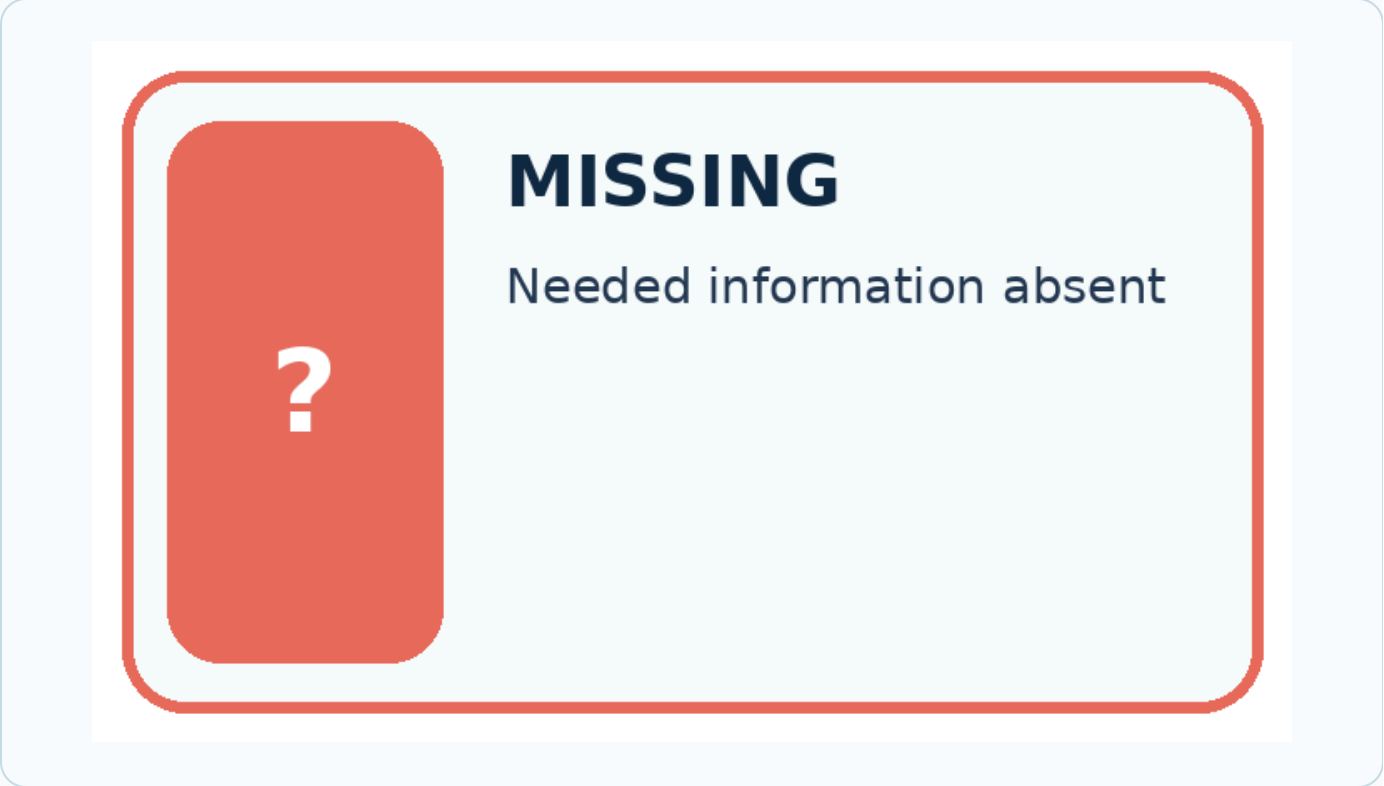
Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Recount an uneven sample, then inspect a separate volunteer poll

> A weighted illustration needs assumptions that arithmetic cannot establish.

Name the three different count records

Keep these examples separate: opening respondents8/10 club and2/10 other; changed-mix exercise72/90 and2/10; pictured comic poll12/15 and3/15, with a separate known grade composition30 club/90 other. Only the pictured scenario supplies grade group sizes for a weighting illustration.

Try the changed-mix proportion

Suppose 90 club respondents give 72 Yes and 10 others give 2 Yes. What is the combined observed proportion?

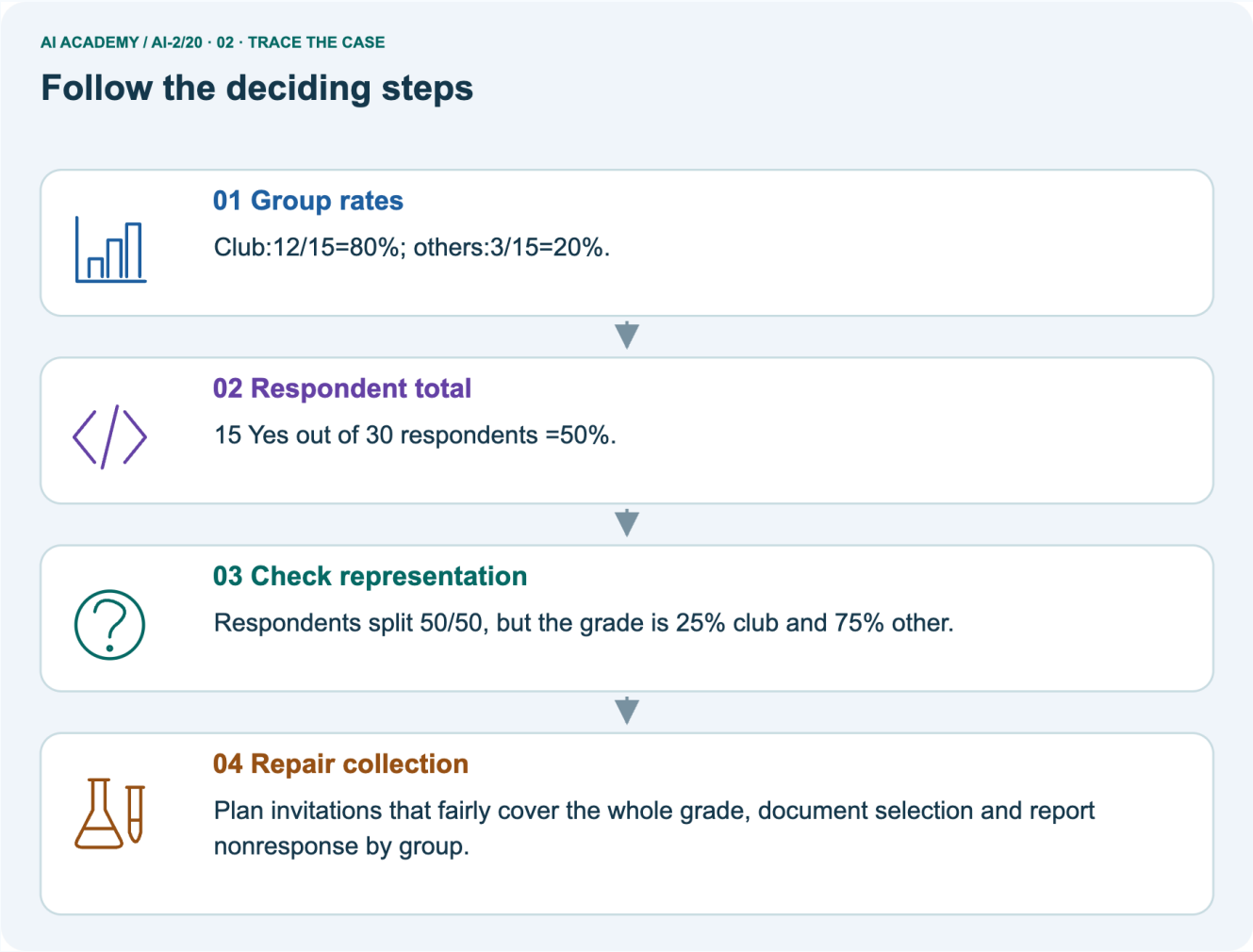
Read the separate comic-poll facts

A fictional school polls learners about adding comic recommendations.

Library-club respondents: 12 Yes out of 15. Other respondents: 3 Yes out of 15.

The full grade has 30 club members and 90 others. These respondents were volunteers, not random selections.

Trace respondent rates and grade representation



Follow the labelled picture with the matching explanation. Open the vector diagram

The combined respondent result gives club members more influence than their share of the grade.

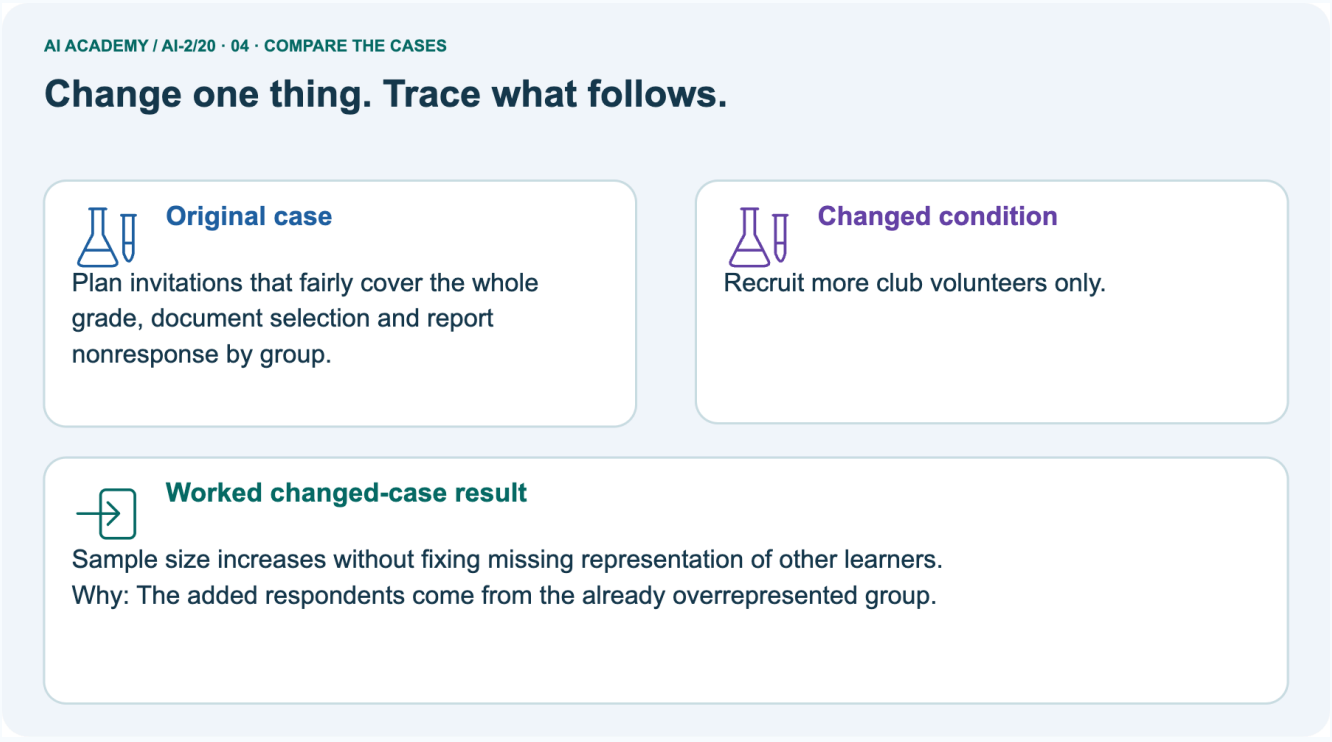
A weighted illustration using 25% and 75% gives 0.25×**80%**+0.75×20%=35%, but volunteers may differ within each group, so weighting alone does not establish the true grade preference.

Improve the collection **process** rather than announcing the weighted number as fact.

Predict more club volunteers only

- Change one thing: Recruit more club volunteers only. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain why35% is an illustration

- Read the labelled picture: Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?

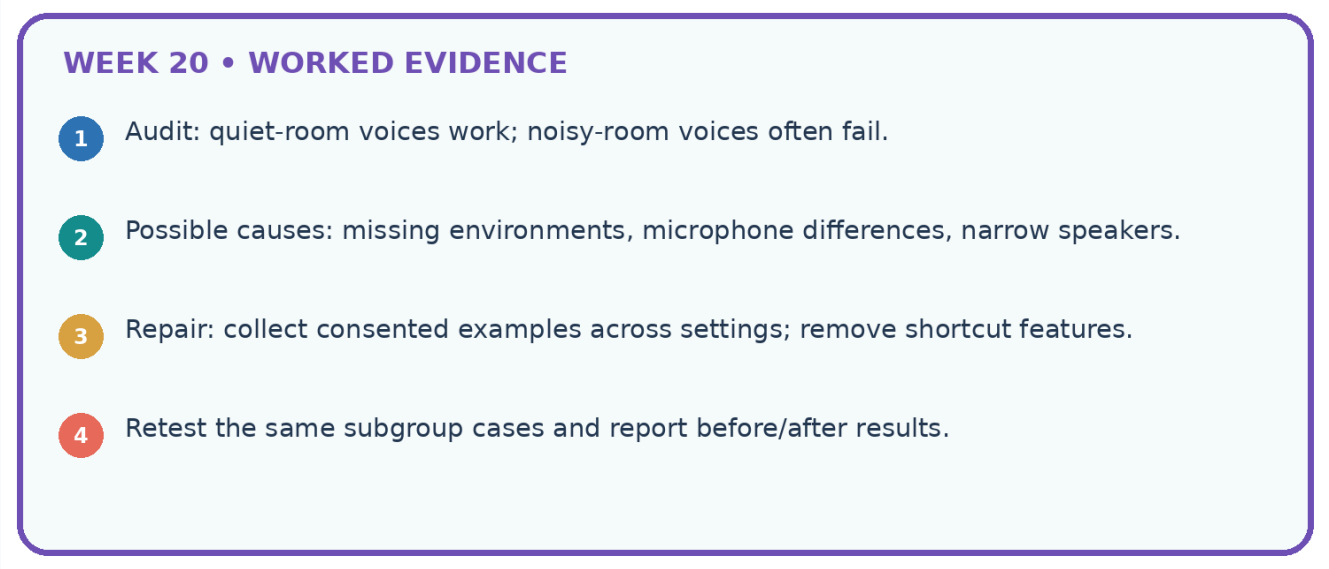
- Neither raw nor weighted volunteer rates prove the whole grade’s opinion; privacy and consent rules still apply.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Audit: quiet-room voices work; noisy-room voices often fail.

2. Possible causes: missing environments, microphone differences, narrow speakers.

3. Repair: collect consented examples across settings; remove shortcut features.

4. **Retest** the same **subgroup** cases and report before/after results.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Repair Bias in Data

The facts we will use

AI ACADEMY / AI-2/20

Volunteer group comparison

	Yes responses	Respondents	Grade members
Club	12	15	30
Others	3	15	90

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/20

Repair Bias in Data

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Group rates

Club:12/15=80%; others:3/15=20%.



2. Respondent total

15 Yes out of 30 respondents =50%.



3. Check representation

Respondents split 50/50, but the grade is 25% club and 75% other.



4. Repair collection

Plan invitations that fairly cover the whole grade, document selection and report nonresponse by group.

CHANGE ONE THING

Recruit more club volunteers only.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Group rates	Club:12/15=80%; others:3/15=20%.
2. Respondent total	15 Yes out of 30 respondents =50%.
3. Check representation	Respondents split 50/50, but the grade is 25% club and 75% other.
4. Repair collection	Plan invitations that fairly cover the whole grade, document selection and report nonresponse by group.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Group differences do not identify a cause

> Keep coverage, small counts and causal explanations distinct.

Limit the group comparison

Group differences suggest that coverage matters, but do not explain why choices differ. Small group counts also make percentages unstable.

Use neutral group labels in fictional data rather than personal or sensitive attributes.

Challenge the causal claim

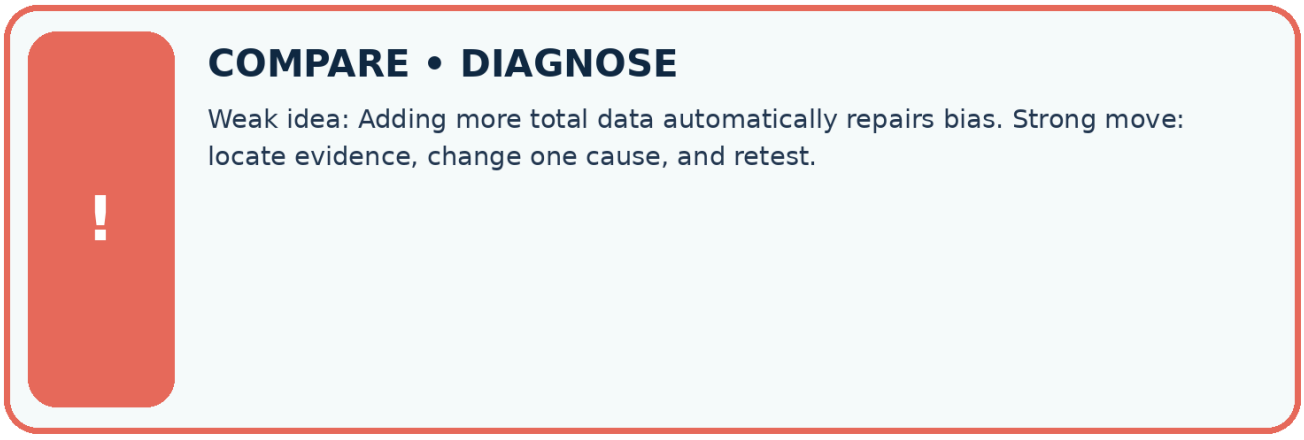
> **Club members favor coding more often. Does club membership necessarily cause that preference?**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.



Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 8

Repair access and selection rather than duplicating records

> A larger version of the same restricted sample can retain the same problem.

Propose a collection repair

Surveying another 100 people from the same undercovered source may make the sample larger without fixing the selection problem. Improve access to missing groups and record **nonresponse**; do not simply duplicate existing rows.

Combine the separate W2 counts

Use workbook W2 for the connected example “**Selection bias** and coverage”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge successful execution proves the claim

- Use the worked case for **Selection bias** and coverage. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the weighted illustration’s limit

Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.

TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; **RETEST** EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Practise with less help: Combine counts

Check your reasoning

Explain the picture

My explanation using a pictured step and the given facts

MAIN LESSON · SECTION 9

Audit a controlled change in respondent mix

> A numerical comparison can expose sensitivity without proving a collection repair worked.

The complete two-scenario count record

Fictional comparison	Club Yes / responses	Other Yes / responses	What is held fixed
A · Equal respondent counts	8/10	2/10	Within-group observed rates80% and20%
B · More club responses	72/90	2/10	The same within-group rates; different respondent mix

Predict, calculate, compare and propose a repair

1. Use these supplied illustrative counts, not a real new survey. Predict both group rates and the combined observed rate for A and B. Execute the opening script for A, then change only club_yes,club_n to72,90 for B; preserve both outputs or label a paper calculation. Explain why the combined rate changes despite unchanged within-group rates. No grade group sizes or random selection are supplied for these two cases, so neither combined rate is established as the grade preference. Compare a collection repair using a whole-grade frame and accessible invitations with merely duplicating a response or recruiting only the existing club. Predict what a repair might improve, but do not invent new response counts or model performance. Keep the independent18/20 and2/10 task unused during practice.

Keep observations and next-step predictions separate

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

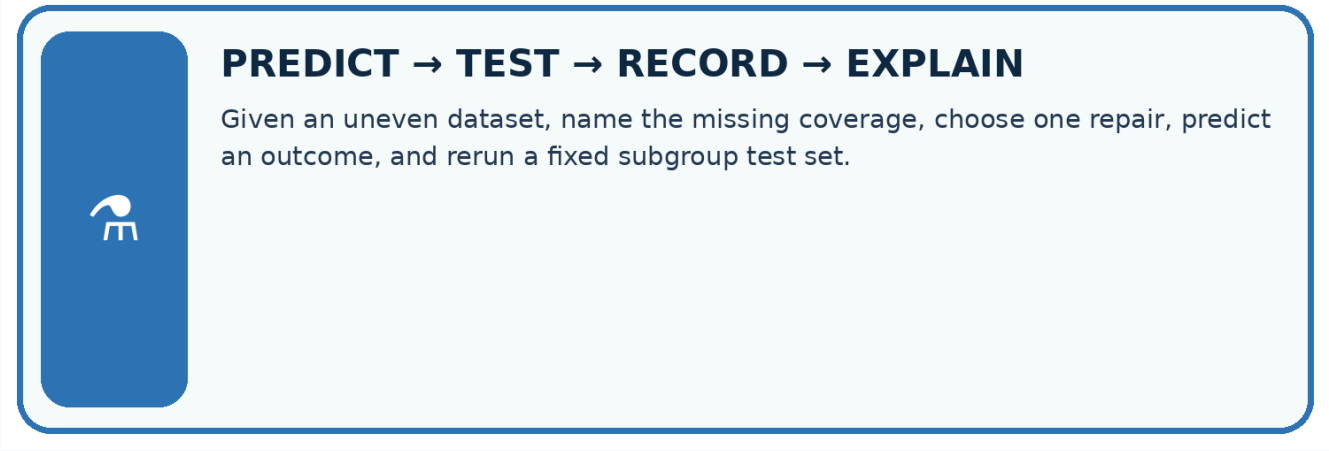
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Use the supplied two-scenario respondent-mix audit. Calculate group and combined rates, explain selection limits and propose a collection repair. No retrained model or **subgroup**-test outputs are supplied; do not describe a proposed repair as an observed improvement.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: audit the new unequal-group survey

> Show both group rates, combined counts and the claim’s limitation.

The independent survey record

A survey has 18 Yes of 20 club members and 2 Yes of 10 others. Compute both group rates and combined rate. Explain why 20/30 does not automatically describe the whole grade.

Your first attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after the attempt

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Audit independently

Week 20 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A dataset has 18 adventure readers and 2 poetry readers. A recommender ignores poetry preferences. What would you inspect and test after a repair?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Change feature coverage or weights only after locating which reader profiles are underserved.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

A catalogue has 30 adventure titles and 2 poetry titles. A recommender rarely suggests poetry. Design a repair and an evaluation that distinguishes catalogue coverage from scoring preference.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Locate the coverage problem before changing recommender weights

> Keep collection, available titles and scoring evidence separate.

This week’s concrete portfolio contribution

For the fictional recommender, investigate catalogue availability, feedback collection and scoring as separate possible causes of an underserved interest. Keep a fixed development request set to compare one documented change at a time; report per-interest counts and denominators. More copies of one poetry title are not varied catalogue evidence, and few poetry suggestions alone do not identify the cause. Reserve fresh final requests and use invented profiles.

Broaden recommender feedback collection and show respondent group coverage before making user-preference claims.

Keep the checkpoint and human decision

THIS WEEK'S CHECKPOINT Change feature coverage or weights only after locating which reader profiles are underserved.

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

> **Write or draw your reasoning and evidence.**

What should remain under **human control**?

> **Write or draw your reasoning and evidence.**

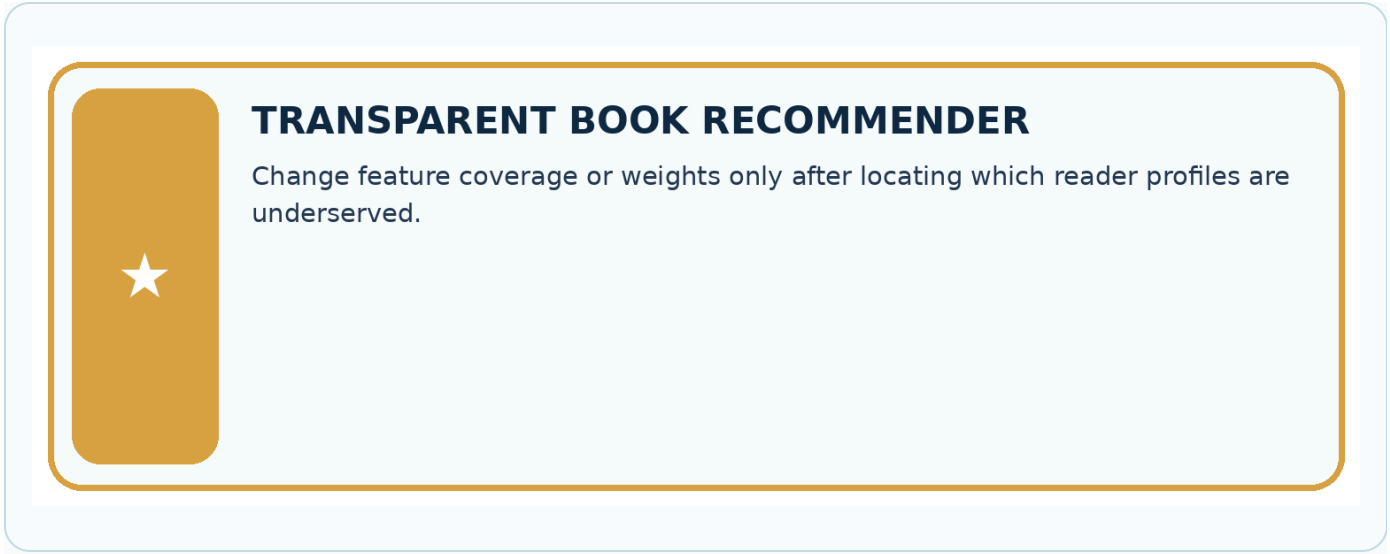
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: name the collection limit beside the calculated rate

> A correct fraction cannot repair who was missing.

Explain without notes

- Close your notes. Explain this goal in your own words: Identify selection and coverage **bias**

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
bias	A systematic tendency in data, decisions or outcomes that can distort a result or disadvantage people.	
rebalance	Change the relative representation of groups or classes in data using a stated method.	
subgroup	A defined subset of cases considered separately in analysis.	
retest	Run a test again after a change or under specified new conditions.	

Terms for the connected explanation

Word	Meaning
Selection bias	Systematic distortion caused by who enters the data
Undercoverage	Leaving part of the target group out of selection

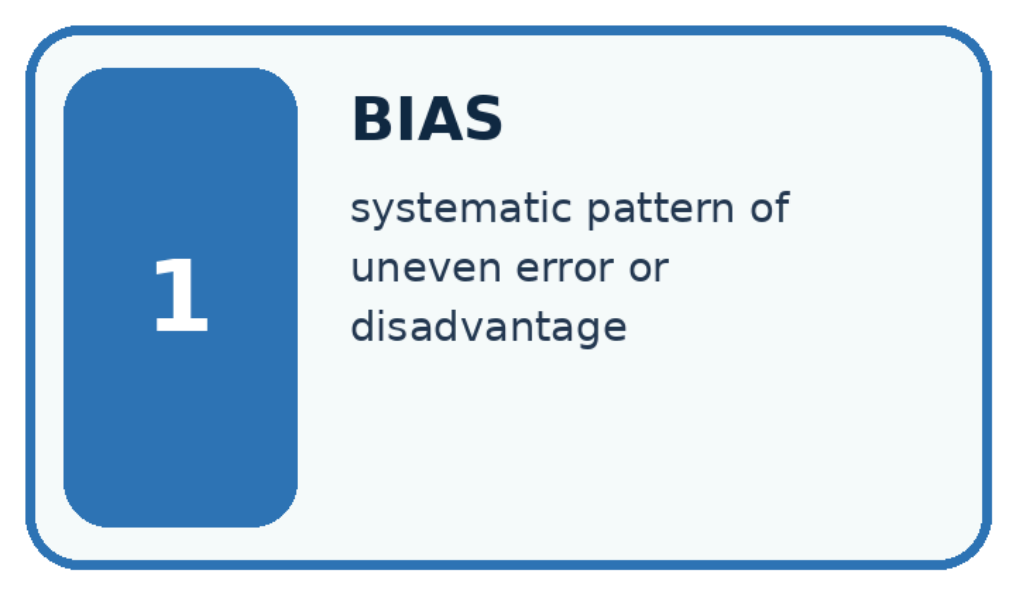
Word	Meaning
Nonresponse	A selected participant does not respond

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<summary>Earlier wording, alternatives and picture transcripts</summary>

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



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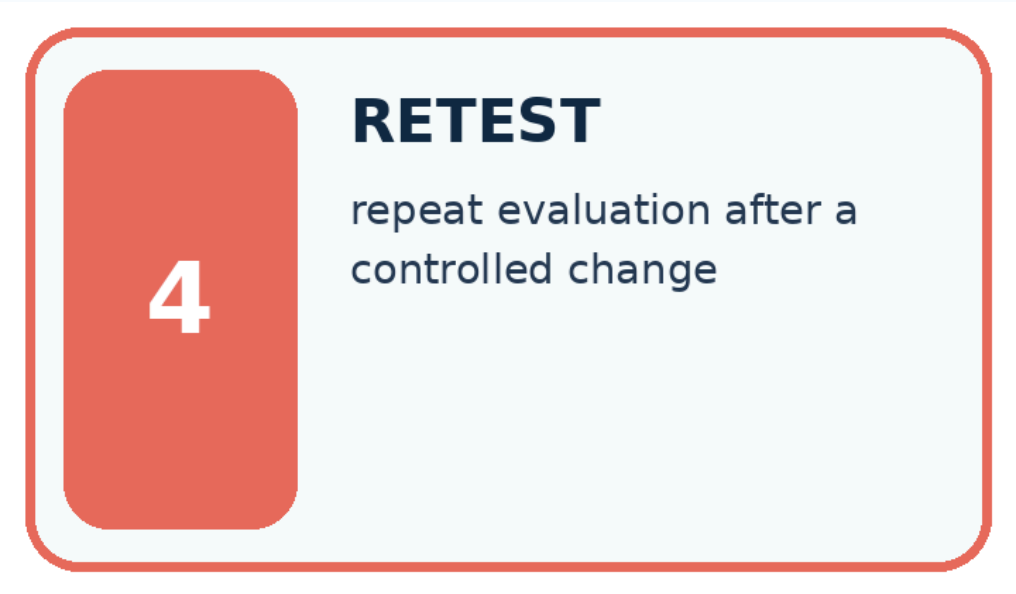
REBALANCE

change coverage across groups or conditions

3

SUBGROUP

meaningful part of the population



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

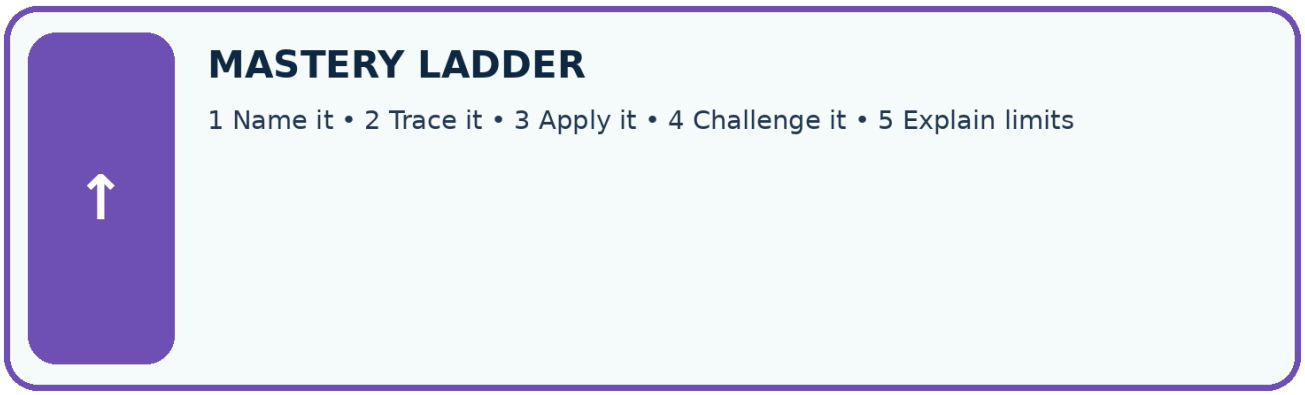
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Audit independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

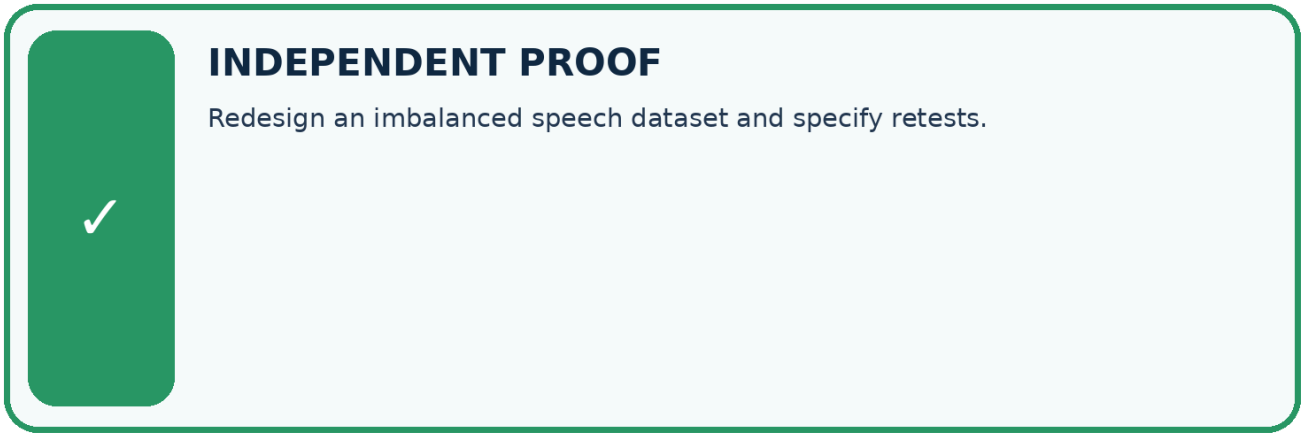
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Audit independently is the selected independent case.



EXPLANATION CHECK Combine observed counts with their denominators, locate coverage and nonresponse limits, and keep causal claims separate.

Write or draw your reasoning and evidence.

▼ **Extra practice: Repair coverage with varied examples**

REFERENCE · SECTION 13

Extra practice: Repair coverage with varied examples

> Try the question before opening its worked explanation.

Read the supplied case and question

Training has eighteen adventure readers and two poetry readers; many intended users prefer poetry.

Propose repair and evaluation.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Collect varied relevant poetry examples with consistent labels. Compare both interests on separate evaluation cases before and after repair.

Evidence or stage	Value or result
Adventure readers	18
Poetry readers	2

▼ Earlier wording, alternatives and picture transcripts

Week 20 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Repair coverage with varied examples

Training has eighteen adventure readers and two poetry readers; many intended users prefer poetry.

Propose repair and evaluation.

Collect varied relevant poetry examples with consistent labels. Compare both interests on separate evaluation cases before and after repair.

▼ **Reference: original lesson checklist**

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Identify selection and coverage bias	Explain the deciding step, show a trace or test, and state one limit.
Compare groups before combining results	Explain the deciding step, show a trace or test, and state one limit.
Explain why more biased data may not help	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional school polls learners about adding comic recommendations.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional school polls learners about adding comic recommendations.
- Which labelled step matches this detail? Plan invitations that fairly cover the whole grade, document selection and report **nonresponse** by group.
- 1. Group rates
- 2. Respondent total
- 3. Repair collection

- 4. Check representation

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Repair collection Plan invitations that fairly cover the whole grade, document selection and report **nonresponse** by group.

The combined respondent result gives club members more influence than their share of the grade.

A weighted illustration using 25% and 75% gives 0.25×**80%**+0.75×20%=35%, but volunteers may differ within each group, so weighting alone does not establish the true grade preference.

Improve the collection **process** rather than announcing the weighted number as fact.

Changed-case question

- Changed-case practice: Recruit more club volunteers only. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future **input**.

- 2. Sample size increases without fixing missing representation of other learners.

- 3. We can decide the result without examining the changed **input** or the procedure.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Sample size increases without fixing missing representation of other learners. The added respondents come from the already overrepresented group.

Repair and save your work

Review Sampling and Representation (ai-2/19). Goal: Define a question and target population. Short practice: Can **80%** response be reported as **80%** of the whole grade responding?. Return to this week and explain what you changed.

Review Noise and Outliers (ai-2/17). Goal: Calculate mean and median. Short practice: If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?.

Return to this week and explain what you changed.

Review Decision Trees (ai-2/14). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Why does changing colour alone not change this tree's **prediction**?.

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

</details>

<details>
<summary>Reference: the six-picture journey</summary>

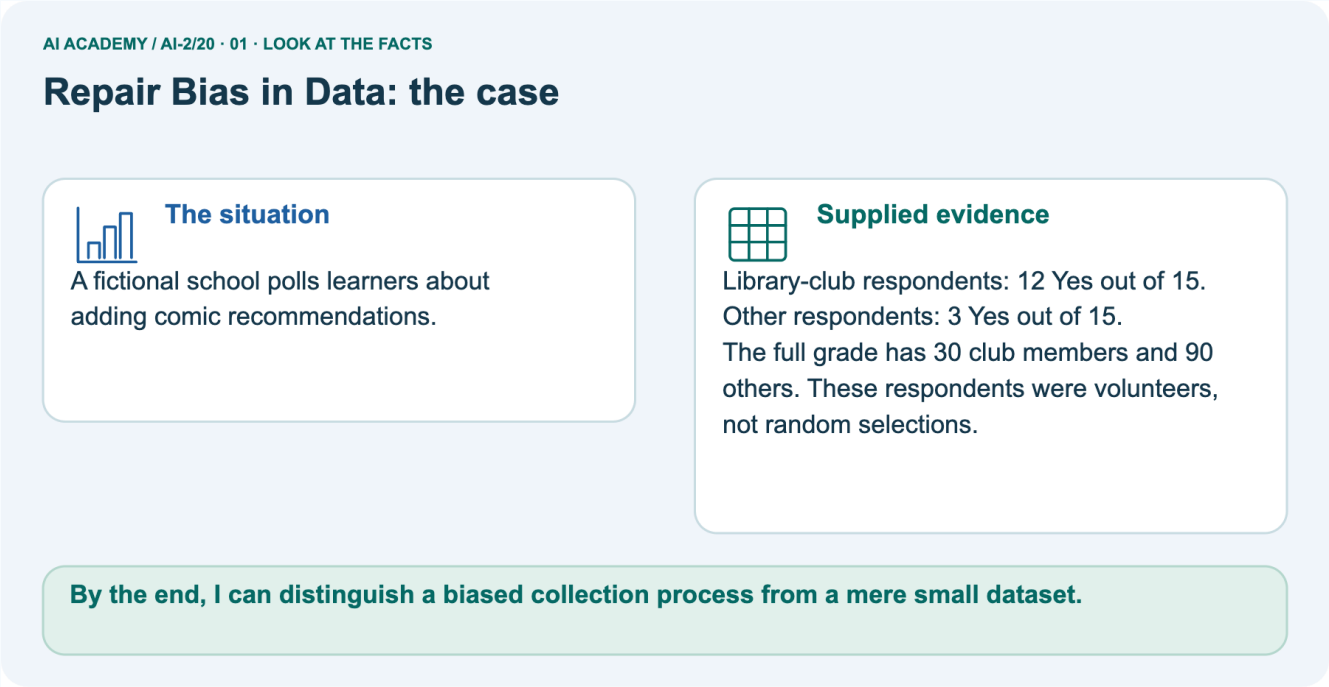
</details>

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Repair Bias in Data: the case



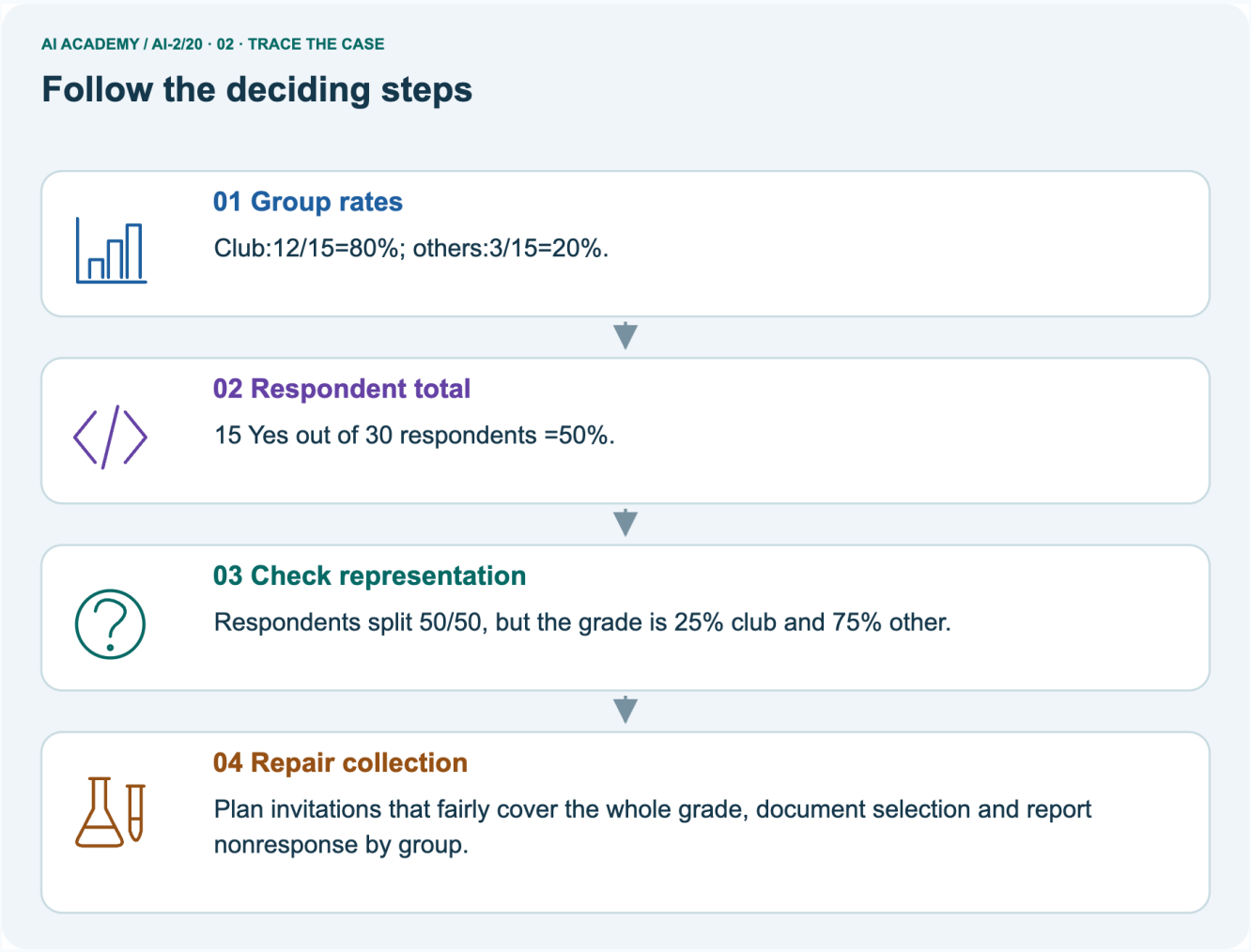
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional school polls learners about adding comic recommendations. Supplied facts: Library-club respondents: 12 Yes out of 15. Other respondents: 3 Yes out of 15. The full grade has 30 club members and 90 others.

These respondents were volunteers, not random selections.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

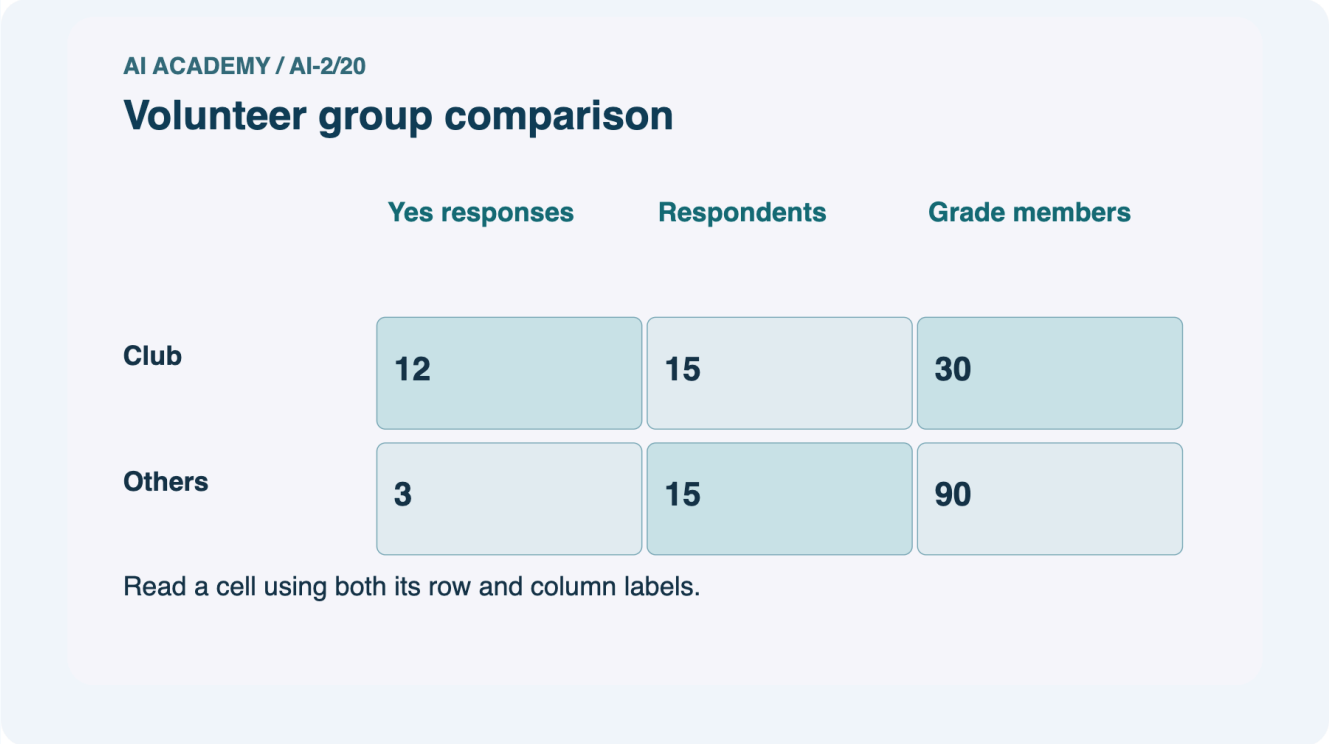


Follow the labelled picture with the matching explanation. Open the vector diagram

Group rates: Club:12/15=**80%**; others:3/15=20%. → Respondent total: 15 Yes out of 30 respondents =50%. → Check representation: Respondents split 50/50, but the grade is 25% club and 75% other. → Repair collection: Plan invitations that fairly cover the whole grade, document selection and report **nonresponse** by group.

Point to each numbered stage and explain which recorded **input** it uses.

03. Volunteer group comparison

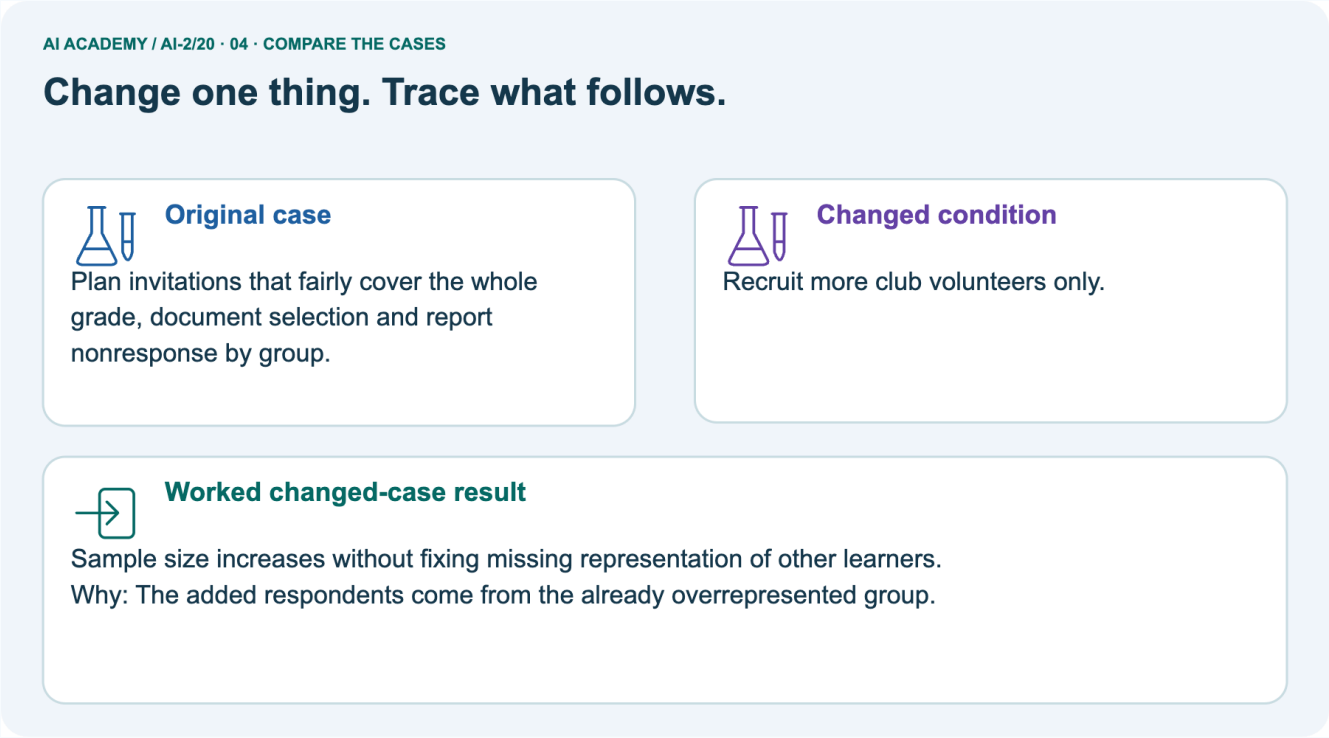


Follow the labelled picture with the matching explanation. Open the vector diagram

Volunteer group comparison. Club: Yes responses = 12, Respondents = 15, Grade members = 30. Others: Yes responses = 3, Respondents = 15, Grade members = 90.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



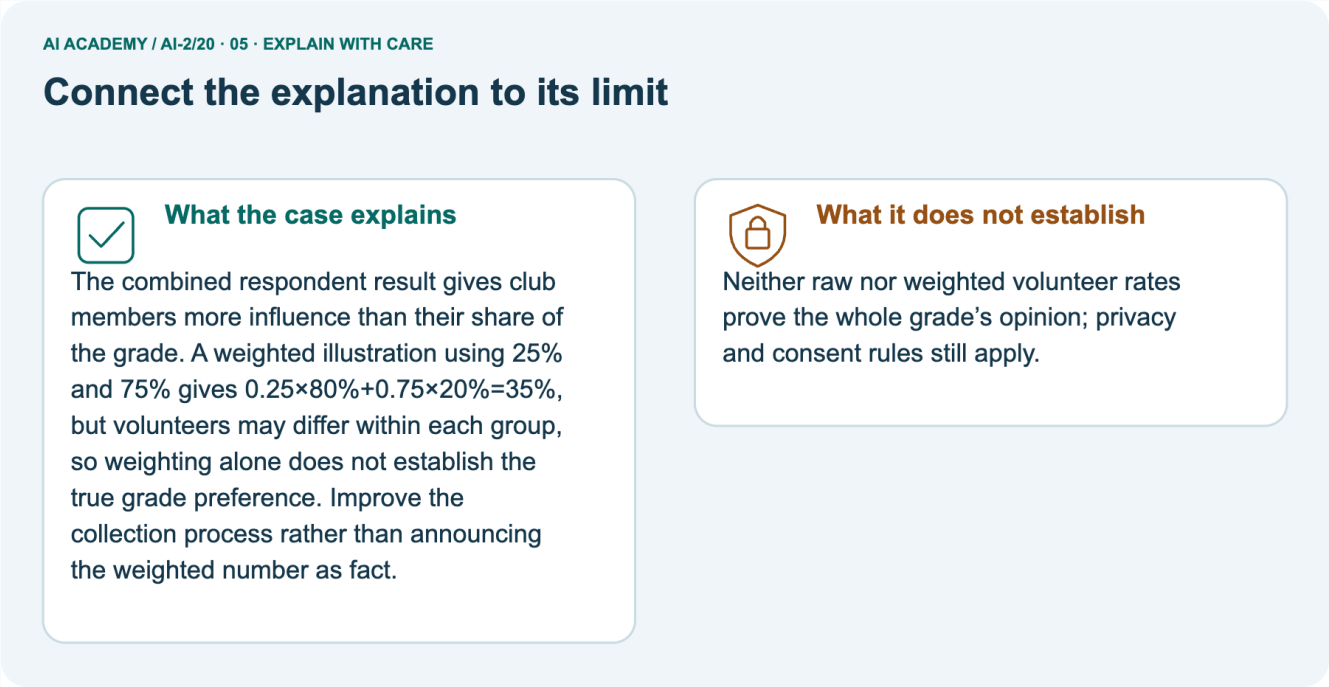
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Plan invitations that fairly cover the whole grade, document selection and report **nonresponse** by group. Changed condition: Recruit more club volunteers only.

Expected result: Sample size increases without fixing missing representation of other learners. Reason: The added respondents come from the already overrepresented group.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

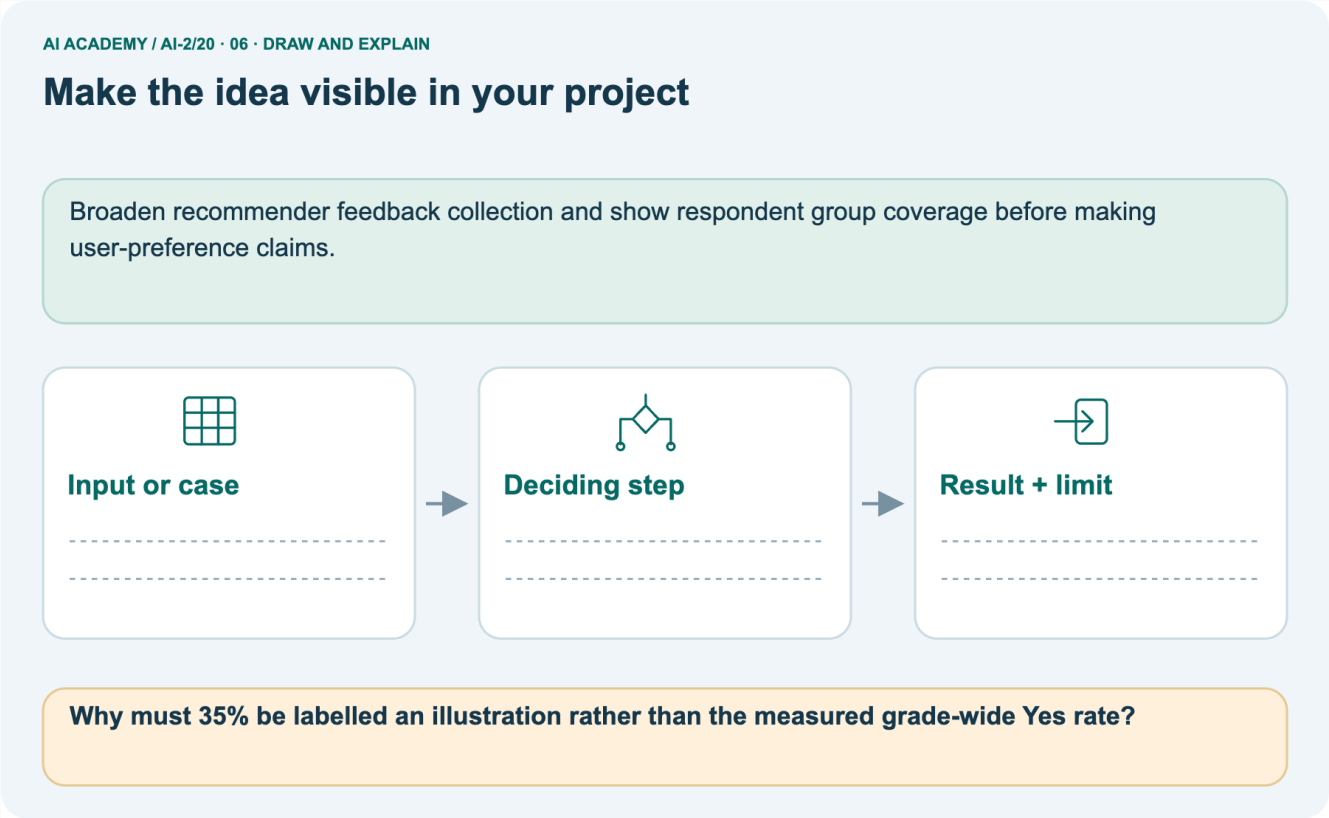
The combined respondent result gives club members more influence than their share of the grade.

A weighted illustration using 25% and 75% gives 0.25×**80%**+0.75×20%=35%, but volunteers may differ within each group, so weighting alone does not establish the true grade preference.

Improve the collection **process** rather than announcing the weighted number as fact. Limit: Neither raw nor weighted volunteer rates prove the whole grade's opinion; privacy and consent rules still apply.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Broaden recommender feedback collection and show respondent group coverage before making user-preference claims. Draw your **input**, deciding step, result and limit.

Supported question: Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Repair Bias in Data: the case

02. Follow the deciding steps

03. Volunteer group comparison

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.